

0.1 Standard Model of Particle Physics

The Standard Model (SM) of particle physics is a framework which contains our current understanding of the matter and three of the four forces found in the universe. Within the SM there are 12 elementary fermions, 5 vector bosons which mediate the three fundamental forces and a scalar boson responsible for particle masses (Figure 1). It is important to note that the SM is not a complete model of the universe. A great example of this is the gravitational force which is not described by the theory, however at the small scales of particle physics this force is neglected due to its insignificant strength when compared to the other three forces (strong, electromagnetic and weak).

Quarks	u Up	c Charm	t Top	g Gluon	H Higgs
	d Down	s Strange	b Bottom	γ Photon	Gauge Bosons
Leptons	e Electron	μ Muon	τ Tau	Z Z Boson	
	ν_e Electron Neutrino	ν_μ Muon Neutrino	ν_τ Tau Neutrino	W W Boson	

Figure 1: The Standard Model of particle physics with the 12 fermions shown in purple and green, the five vector bosons shown in red and the Higgs scalar boson shown in yellow.

The SM comprises of a quantum field theory (QFT) and is based on the non-Abelian gauge symmetry groups $SU(3) \times SU(2) \times U(1)$. The $SU(3)$ governs the strong interactions and the $SU(2) \times U(1)$ the electroweak interactions. Each group contains their respective gauge bosons: gluon for the strong and W, Z and photon for the electroweak. The $SU(3)$ is an exact gauge symmetry where as the electroweak symmetry is spontaneously broken by the Higgs field, resulting in the massive electroweak bosons ($W^\pm$ and $Z^\pm$). Due to this broken symmetry which gives rise to massive

quarks, the strong and weak eigenstates diverge. The weak eigenstates (d' , s' , b') of down, strange and bottom quarks which are associated to the weak interaction are related to the mass eigenstates (d , s , b) through the Cabibbo-Kobayashi-Maskawa (CKM) matrix V_{CKM} [1, 2].

$$\begin{bmatrix} d' \\ s' \\ b' \end{bmatrix} = V_{\text{CKM}} \begin{bmatrix} d \\ s \\ b \end{bmatrix} = \begin{bmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{bmatrix} \begin{bmatrix} d \\ s \\ b \end{bmatrix} \quad (0.1)$$

Utilising the elements of the CKM matrix it is possible to calculate the probability that a quark type i will transition into type j with $|V_{ij}|^2$.

For a unitary 3×3 matrix the CKM can be reduced from nine parameters to four: three mixing angles (θ_{12} , θ_{13} , θ_{23}) which dictate the couplings between each generation of quarks and one phase (δ) which introduces CP violation [2]. A number of different parametrisations are possible but the standard choice is [3],

$$V_{\text{CKM}} = \begin{bmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{bmatrix} \quad (0.2)$$

where $c_{ij} = \cos \theta_{ij}$ and $s_{ij} = \sin \theta_{ij}$. The mixing angles θ_{ij} can be selected to satisfy $0 \leq \theta_{ij} \leq \frac{\pi}{2}$ such that $c_{ij}, s_{ij} \geq 0$.

Through experiments it is known that there is hierarchical structure to the components of the CKM matrix,

$$s_{12} \ll s_{23} \ll s_{13} \quad (0.3)$$

and it is useful to further parametrise through the Wolfenstein parametrisation [4] (Equation 0.5). For this parametrisation the following definitions are used,

$$\begin{aligned} s_{12} &= \lambda \\ s_{23} &= A\lambda^2 \\ s_{13} &= A\lambda^3(\rho + i\eta) \end{aligned} \quad (0.4)$$

resulting in,

$$V_{\text{CKM}} = \begin{bmatrix} 1 - \frac{\lambda^2}{2} & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1 - \frac{\lambda^2}{2} & A\lambda^2 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^2 & 1 \end{bmatrix} + O(\lambda^4). \quad (0.5)$$

Reducing the CKM to $O(\lambda^3)$ results in the loss of unity. However, unity can be restored by including $O(\lambda^5)$ [5] correction to the bottom left component of Equation 0.5,

$$V_{td} = A\lambda^3(1 - \bar{\rho} - i\bar{\eta}) \quad (0.6)$$

where

$$\begin{aligned} \bar{\rho} &= \rho(1 - \frac{\lambda^2}{2}) \\ \bar{\eta} &= \eta(1 - \frac{\lambda^2}{2}) \end{aligned} \quad (0.7)$$

The unitary nature of the CKM matrix imposes conditions on the row and columns

$$\begin{aligned} \sum_i V_{ij}V_{ik}^* &= \delta_{jk} \\ \sum_j V_{ij}V_{kj}^* &= \delta_{ik} \end{aligned} \quad (0.8)$$

where δ_{jk} denotes the Kronecker delta function, which equals 1 when $j = k$ and 0 otherwise. This results in six vanishing combinations of the CKM matrix which can be represented as triangles, with the most common being

$$V_{ud}V_{ub}^* + V_{cd}V_{cb}^* + V_{td}V_{tb}^* = 0 \quad (0.9)$$

and can be modified with the Wolfenstein parametrisation (Equation 0.10) which is depicted in Figure 2.

$$1 + \frac{V_{td}V_{tb}^*}{V_{cd}V_{cb}^*} = -\frac{V_{ud}V_{ub}^*}{V_{cd}V_{cb}^*} = \bar{\rho} + i\bar{\eta} \quad (0.10)$$

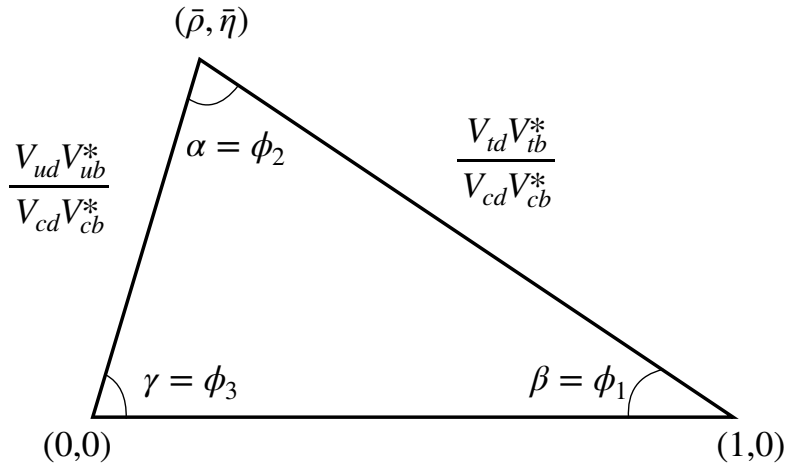


Figure 2: Sketch of the most common unitary triangle $V_{ud}V_{ub}^* + V_{cd}V_{cb}^* + V_{td}V_{tb}^* = 0$.

The CKM matrix components are fundamental parameters of the SM. Through a large number of experiments the elements of the CKM matrix can be measured and are combined to form a global fit [6]:

$$\begin{bmatrix} |V_{ud}| & |V_{us}| & |V_{ub}| \\ |V_{cd}| & |V_{cs}| & |V_{cb}| \\ |V_{td}| & |V_{ts}| & |V_{tb}| \end{bmatrix} = \begin{bmatrix} 0.97436^{+0.00014}_{-0.00019} & 0.22498^{+0.00081}_{-0.00062} & 0.00373^{+0.00019}_{-0.00015} \\ 0.22484^{+0.00081}_{-0.00061} & 0.97351^{+0.00016}_{-0.00019} & 0.04160^{+0.00066}_{-0.00121} \\ 0.00857^{+0.00018}_{-0.00037} & 0.04088^{+0.00066}_{-0.00119} & 0.999125^{+0.000052}_{-0.000027} \end{bmatrix} \quad (0.11)$$

A feature of the CKM matrix is that the diagonal components which describe the transitions within the quark generations dominate, compared to the off-diagonal elements which are suppressed for transitions between one generation and doubly suppressed for transitions between two generations. This suppression between two generations produces rare decays, such as $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ described in Section .

Through measurement of the CKM matrix components it is possible to constrain the unitary triangle (Figure 2) shown in Figure

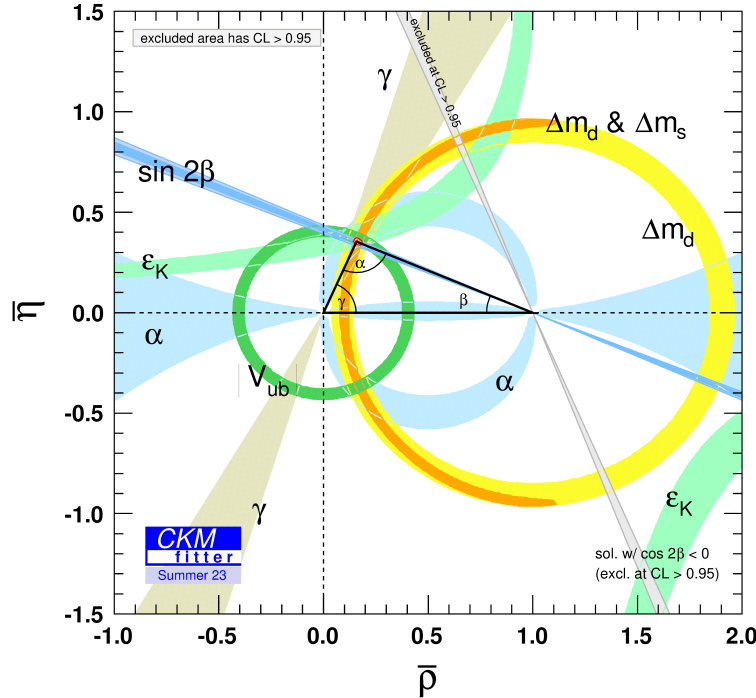


Figure 3: Constraints for the most common unitary triangle in the $(\bar{\rho}, \bar{\eta})$ plane [6].

Including the four CKM components discussed above there are nineteen free parameters of the SM:

- Four CKM parameters discussed above.
- Six Quark masses (Up, down, charm, strange, top and bottom).
- Three Lepton masses (Electron, muon and tau).
- Three Gauge Couplings (U(1), SU(2) and SU(3))
- Two Higgs parameters (Vacuum expectation value and mass).
- Quantum chromodynamics (QCD) vacuum angle.

Each one of these parameters needs to be measured through experiment to the highest precision so that any discrepancies can be resolved. Any deviations from theoretical understanding could be hint of physics beyond the Standard Model (BSM).

0.2 Kaon Physics

The first possible observation a heavy meson was in 1943 by Leprince-Ringuet and L'heritier [7, 8] and predated even the discovery of the pion in 1947 [9]. Utilising a cloud chamber to measure cosmic rays they found the first evidence of a particle with a mass of 990 times the mass of the electron (0.511 MeV [10]). However, it was possible that the upper limit on the measured mass could reach the proton mass so they were not credited with the discovery. In 1947 while similarly studying cosmic rays with cloud chambers G. D. Rochester and C. C. Butler published a result which included two pictures showing forked tracks [11]. One of the tracks proved to show the decay of a neutral particle into two charged daughter particles, which turned out to be decay of the neutral kaon to two pions. The other track showed the decay of a charged particle into charged and neutral daughter particles. This result was later confirmed in 1950 by a group in Cal-Tech who observed multiple instances of these decays [12]. However, between these publications a group in Bristol which included physicist Rosemary Brown, used emulsion chambers to observe the decay of a charged particle into three charged particles [13], known then as the tau decay. This resulted in what was known as the tau-theta problem, where the seemingly the same particle was decaying into two different final states with different parities:

Theta to two pions (+1) and Tau to three pions (-1). This eventually led to the discovery that parity is not conserved in weak interactions [14].

In the early 1950s these particles would be described as K-mesons, for which there are four types (K^+ , K^- , K^0 and $\bar{K}^0$). Experiments continued to measure the characteristics of the K mesons and found features such as long lifetimes (10^{-10} seconds) compared to the fact their production was much faster [15]. These deviations from the other mesons, mainly the pion suggested the existence of a new quantum number named strangeness [16, 17]. The strange quantum number applies to all strongly interacting particle and is conserved in strong interactions but violated in weak interactions.

Into the latter 1950s and early 1960s the neutral kaon became the center of number of studies. Despite the fact that parity is violated in weak interactions it was still believed that the combination of charge-parity (CP) would be conserved. Through experiment the two weak eigenstates of the neutral kaon showed to have different lifetimes, which resulted in their names K_L and K_S [18]. Each of these eigenstates had different CP values with the K_S with $CP = +1$ decaying to two pions and the K_L with $CP = -1$ observed decaying to three pions. As CP was considered to be conserved the physical eigenstates would be,

$$\begin{aligned} |K_1^0\rangle &= \frac{1}{\sqrt{2}} (|K^0\rangle + |\bar{K}^0\rangle) \\ |K_2^0\rangle &= \frac{1}{\sqrt{2}} (|K^0\rangle - |\bar{K}^0\rangle) \end{aligned} \tag{0.12}$$

where K_1^0 has a $CP = +1$ and K_2^0 has $CP = -1$. However, in 1964 it was observed that the K_L decayed to two pions,

0.2.1 $K^+ \rightarrow \pi^+ \nu \bar{\nu}$

0.2.2 $K_L^0 \rightarrow \pi^0 l^+ l^-$

0.2.3 $K_L^0 \rightarrow \pi^0 \nu \bar{\nu}$

0.3 Dark Photon

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